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SALES FORECAST AND PRICE OPTIMISATION ANALYSIS FOR YKS RESTAURANT

ABSTRACT

To project sales trends over the three months (April, May, June) and implement pricing optimizations aligned with the menu engineering principles of YOKOSO Restaurant, with the goal of maximizing revenue and enhancing the overall dining experience for patrons in Pune.

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DATA

We collected the historical sales by item transactions data from the POS system of YKS restaurant. In which the records are spanning from January 2023 up till march 2024.

The data contained nearly about 49000 records The selected time period, spanning from January 2023 to March 2024(14 months), was chosen to capture seasonal variations and long-term trends in sales performance from past one year. Based on this Data we forecasted the sales of three months April 2024, May 2024, June 2024.

METHODS

THE METHODS WE USED FOR ENTIRE ANALYSIS INCLUDES:

Data Analysis: Utilized Various Plots: Visualized data using histograms, scatter plots, and line charts to explore trends, distributions, and relationships.

Applied Statistical Techniques: Employed descriptive statistics and hypothesis testing to validate assumptions and ensure analysis reliability.

Menu Analysis: profitability of menu items by considering their contribution margins (selling price minus food cost). Classified items into categories based on profitability.

Popularity Assessment: Determined the popularity of menu items by analysing sales data. Identified best-selling and low-selling items to understand customer preferences.

Conducted Exploratory Data Analysis (EDA): Explored data thoroughly to uncover patterns and anomalies, facilitating hypothesis generation.

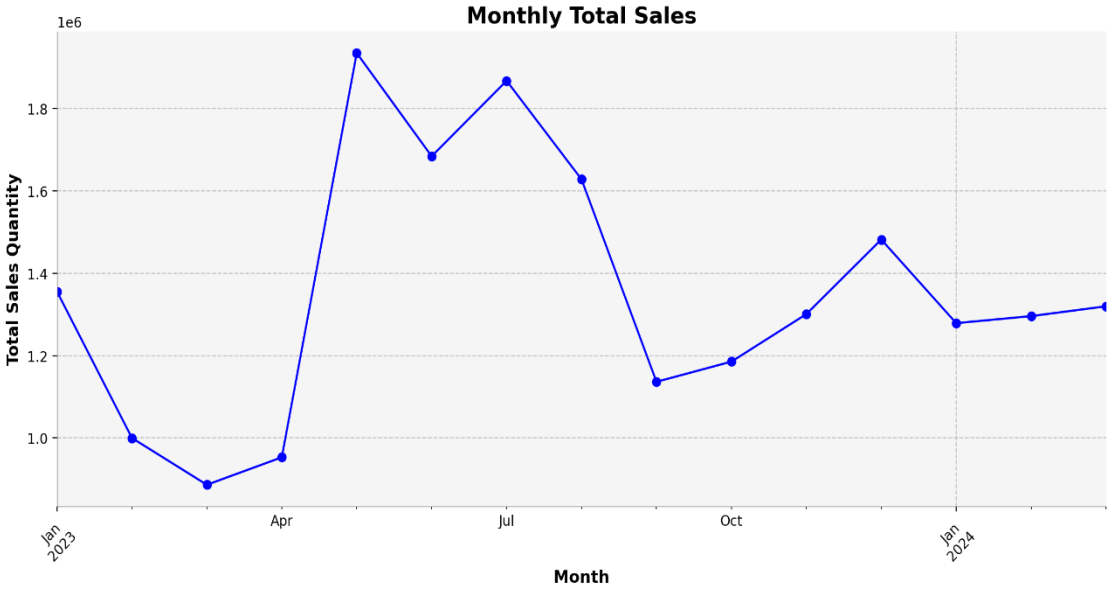
Forecasting Methods:

- Regression Analysis: Used linear regression, decision tree, and random forest algorithms to predict sales by analysing historical data.
- Time Series Analysis: Applied ARIMA and LSTM models to capture temporal patterns and forecast future sales based on seasonality, trends, and irregular components.

INSIGHTS 🎯

Monthly Sales

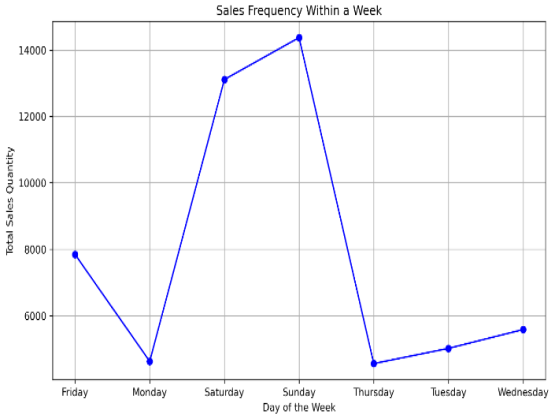
Talking about the monthly sales it is seen that there is peak in the month of May it somehow remained consistent till month of July and August and it drastically dropped in the month of September.



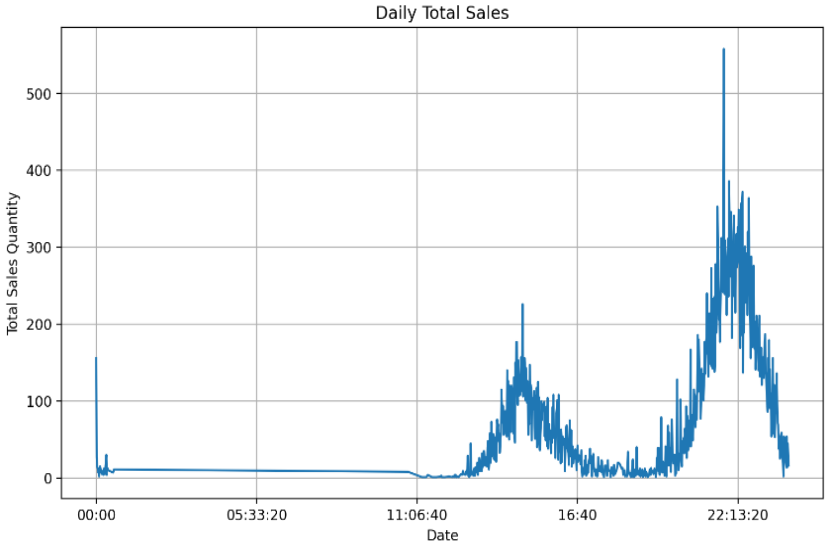
Has restaurant made any changes in this particular time period with respect to Menu or pricing or both? If yes what are those changes. And if no has quality or quantity been compromised after that, or any staff related changes made. And if no what are the other thing that can might affect the Sales.

Weekly Sales Frequency:

Here, weekly data is quite predictable that most of the sales are happening in weekends as the most of the target customers are from corporate society.



Peak Hours

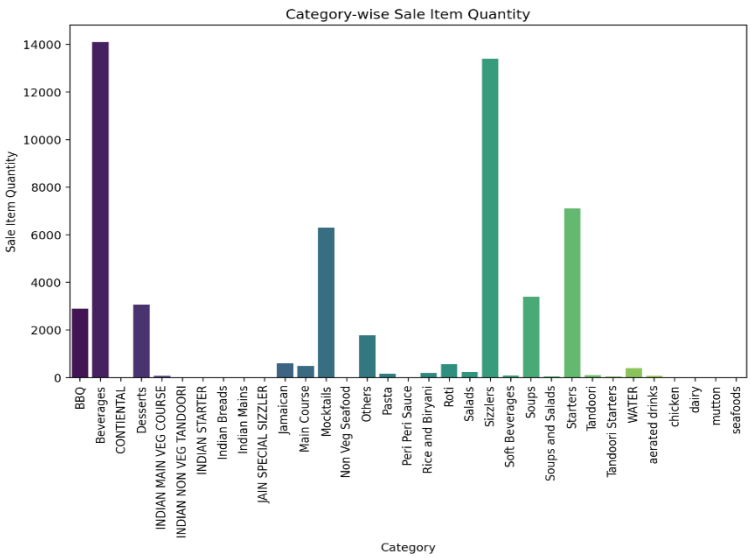


By looking at the graph of daily sales timings here it can be clearly seen that there are only two peaks in the Sales throughout the day that timings in afternoon between 2pm to 4 pm And in the evening, it is 7 pm to 11 pm which is obviously a dinner time. and it is observed in weekdays there are very less sales as compared to weekends Friday, Saturday, Sundays, if the restaurant wants customers in weekdays also, Restaurant can offer discounts on particular weekdays to attract customers.

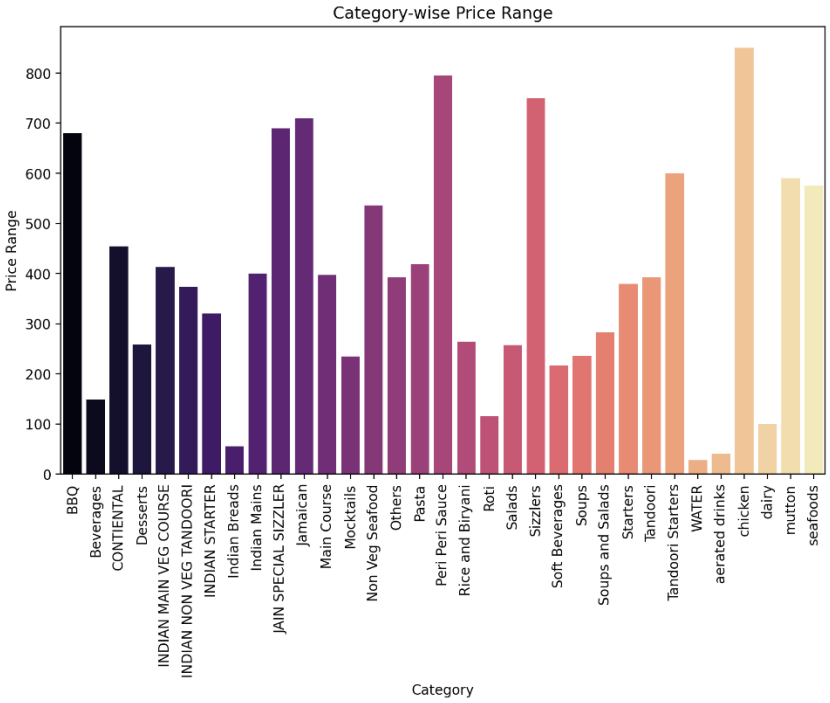
Top quantity sold Per Category:

Top Selling categories:

- Sizzlers
- Beverages
- Starters
- Mocktail
- Soups
- Desserts

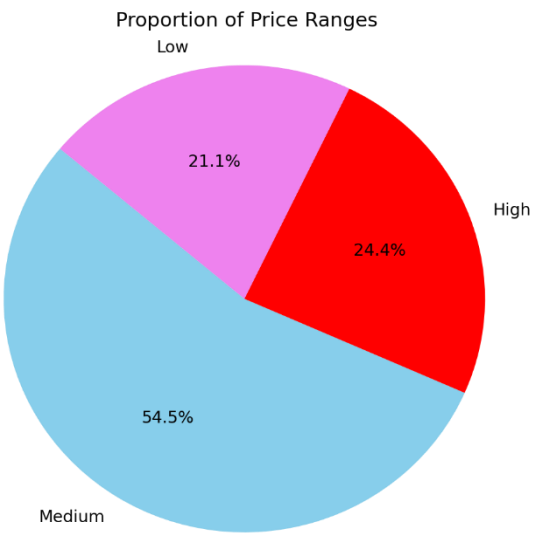


Price Range of different categories:



Here in this visual menu prices of most popular categories are clustered in between the range of 500 to 800 for the most prominent categories. Suggesting that prices can be tweaked/hiked for high demanding item. And be carefully optimised for the supporting categories like desserts, starters, soups, and water.

We had divided the whole price distribution into three categories Low, medium, High using the summary statistics of Unit Rates of items. The following pie chart is indicating the proportion of each price range.



Price Range	Category
0 - 230	Low
231 -670	Medium
670 and above	High

Category-wise Top Items

TOP SIZZLERS	UNIT RATES
VEGETABLE SIZZLER WITH CORN CHEESE CUTLET	690
VEGETABLE CHINATOWN	720
MIX VEG CUTLET	750
VEGETABLE MIX GRILL	700
CHICKEN SHASHLIK	740
CHICKEN CHINATOWN	780
MALAYSIAN GRILLED CHICKEN	740
PANEER SATELLITE	670
FISH SIZZLER SURMAI	730
SEAFOOD MIXED GRILL	850

TOP BEVERAGES	UNIT RATES
MINERAL WATER	50
FRESH LIME SODA	110
AERATED DRINKS	80
DIET AERATED DRINKS	100
PEACH ICED TEA	150
LEMON ICED TEA	150
TEA / COFFEE	20
FRUIT JUICE	120
SODA BOTTLE 750 ML	100
COLD COFFEE	160

TOP STARTERS	UNIT RATES
GARLIC BREAD WITH CHEESE	250
VEG SPRING ROLL	200
CORN CHEESE NUGGETS	290
FRENCH FRIES	220
SHANGHAI CHICKEN	430
VEG CRISPY	200
SHANGHAI PANEER	400
CHICKEN LOLLIPOP	330
MASALA PAPAD	60
CORN CHEESE BALLS	290

TOP MOCKTAILS	UNIT RATES
ROYALTY OF YOKOSO	230
VIRGIN MOJITO	210
VIRGIN PINA COLADA	210
FRENCH KISS	210
BOO BOO SPECIAL	210
MELON WIVES	210
RED WINE SANGRIA	230
GARDEN COOLER	210
KIWI COOLER	210
PASSIONATE	210

TOP SOUPS	UNIT RATES
VEG MANCHOW SOUP	220
ONION AND CHEESE SOUP	190
VEGETABLE LEMON CORIANDER SOUP	220
CHICKEN MANCHOW SOUP	240
CREAM OF CHICKEN	220
MINESTRONE SOUP	190
TOMATO CLASSIC CREAMY SOUP	190
VEG SPICY THAI KOREAN SOUP	210
VEGETABLE SWEET CORN	190
VEG TOM YUM SOUP	200

TOP BBQ	UNIT RATES
BBQ PANEER SQUARE	690
BBQ CHICKEN SQUARE	750
BBQ CHICKEN	770
BBQ EXOTIC VEGETABLES	710
BBQ TIGER PRAWNS	820
BBQ SIZZLING CHICKEN	730
BBQ LAMB FILLET	780
BBQ MANCHURIAN	650
BBQ FISH	800
BBQ PANEER	670

TOP DESSERTS	UNIT RATES
SIZZLING BROWNIE WITH ICE C REAM	300
CARAMEL CUSTARD	250
TIRAMISU	220
FRIED ICE CREAM	270
MALAI KULFI	200
SIZZLING APPLE WITH ICE CREA M	300
BROWNIE WITH ICE CREAM	270
APPLE PIE WITH ICE CREAM	280
VANILLA ICE CREAM	160
HONEY NOODLES WITH ICE CRE AM	280

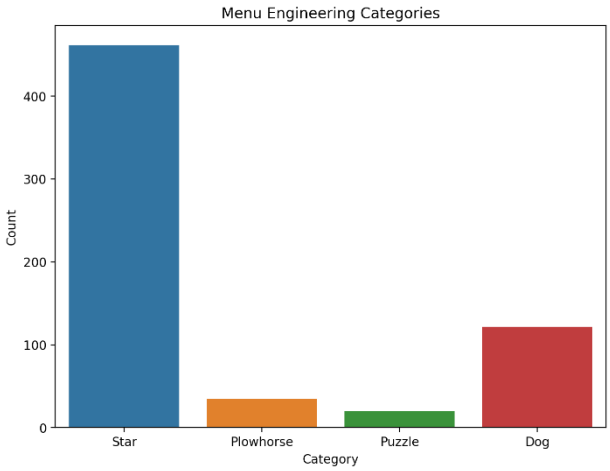
TOP OTHERS	UNIT RATES
THE SOKUDO WOK	660
BUTTER ROTI	70
VEG FRIED RICE	300
PARTY PACKAGE	800
CHICKEN FRIED RICE/NOODLES	320
PENNE PASTA	350
BUTTER ROTI	40
EXTRA CHEESE	50
VEG FRIED RICE\NOODLES	280

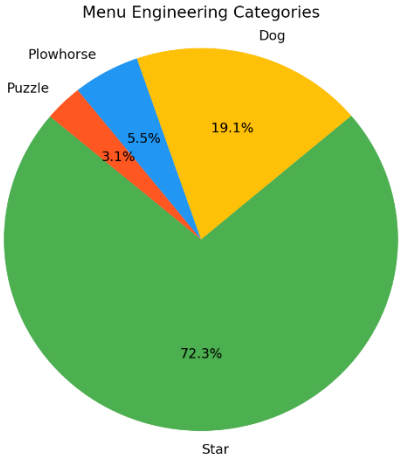
MENU ANALYSIS 🍽

Here we categorized menu item according to criteria required for menu engineering principles.

Menu Item Classification:

- **Stars**: This category consists menu items with High-profit and high-popularity. Which actually Promote and highlight the items to maximize revenue.
- **Plowhorses**: This category consists High-popularity but low-profit items. Consider adjusting prices or portion sizes to improve profitability.
- **Puzzles**: High profitability and low popularity Evaluate whether to keep, modify, or remove these items from the menu.
- **Dogs**: Low-profit, low-popularity items. Consider removing or repositioning these items to optimize menu offerings.





From above Graphical analysis we already have 72.3% menu items in the Star category where we can make some significant price changes according to the desired profit we want to get. Along with that there are 19.1% menu items comes under the Dog category indicating the low profitable and low popular items. And 5.5% item falls under category of Plowhorse and 3.1% items falls under the category of Puzzles.

List of top items in Star Category:

Here in this table, we have categorised top ten items by Total profit which comes under the high profitable and high popular categories where restaurant can make significant price change. can increase the prices for high demanding dishes.

Item Name	Sale Item Qty	Total Profit	Category
Bbq Paneer Square	1015	140106.0	Star
Vegetable Chinatown	946	138374.0	Star
able Sizzler With Corn Cheese	957	133124.0	Star
Bbq Chicken Square	636	95432.0	Star
Mix Veg Cutlet	605	91950.0	Star
Bbq Chicken	547	84238.0	Star
Sizzling Brownie With Ice Crea	1255	78042.0	Star
Mineral Water	7745	77450.0	Star
Vegetable Mix Grill	495	73342.0	Star
Chicken Shashlik	462	70336.0	Star

List of top items in Plowhorses Category :

Item Name	Sale Item Qty	Total Profit	Category
Paneer Lazabab	4	216.0	Plowhorse
Jeera Rice (Half)	12	216.0	Plowhorse
Egg Omelet	7	210.0	Plowhorse
Chicken Lawabdar	3	210.0	Plowhorse
Chicken Hangama Kebab	3	198.0	Plowhorse
CHEESE CHERRY PINEAPPLE	4	190.0	Plowhorse
Santiago Tini	4	184.0	Plowhorse
Grilled Chicken Salad	3	180.0	Plowhorse
Butter Milk	10	180.0	Plowhorse
Paneer Masala	3	180.0	Plowhorse

List of top items in Puzzle Category:

Item Name	Sale Item Qty	Total Profit	Category
Family nonveg combo for 4	2	679.5999999999999	Puzzle
Family combo for 4	2	599.5999999999999	Puzzle
DECORATION	1	400.0	Puzzle
Fish Surami	2	336.0	Puzzle
Peri Peri Lamb Square	2	318.0	Puzzle
FFISH POMFRET TANDOORI	2	312.0	Puzzle
lalaysian Grilled Chicken Sala	2	304.0	Puzzle
ledallion Grilled Chicken Sala	2	296.0	Puzzle
Jamaican Chicken Salami	2	280.0	Puzzle
Spicy Chicken Salami	2	274.0	Puzzle

List of top items in Dog Category :

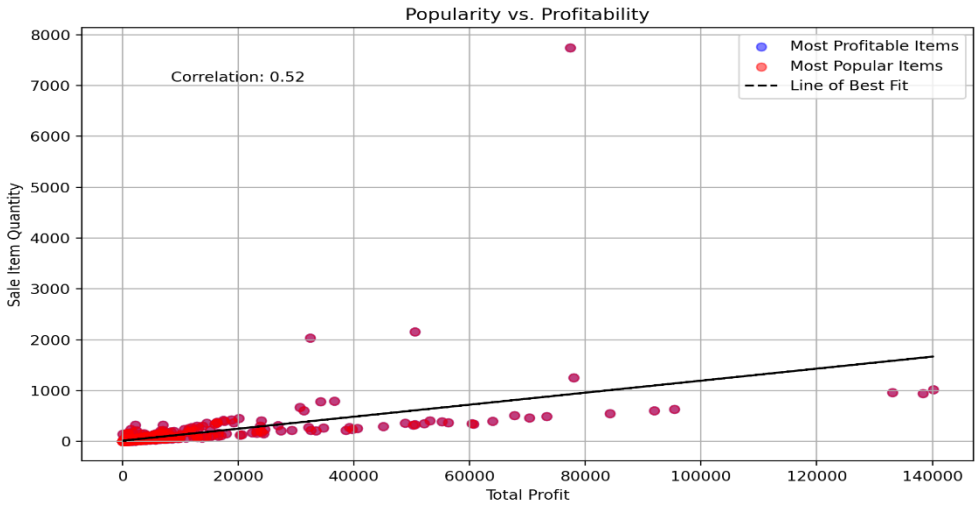
Item Name	Sale Item Qty	Total Profit	Category
DRUMSTICK JUNGLEE KEBAE	2	196.0	Dog
Mutton Biryani	2	180.0	Dog
Chicken Handi	2	180.0	Dog
Shanghai Fried Chicken	2	180.0	Dog
Malaysian King Fish	1	180.0	Dog
Schezwan Style Lamb	1	176.0	Dog
Tiger Prawns	1	174.0	Dog
Mutton Handi (Full)	1	168.0	Dog
Chicken Tangdi	2	164.0	Dog
Peri Peri Lamb Fillet	1	162.0	Dog

Menu analysis data:

the following link contain the menu data with the proper labels of Prices Range categories and Menu engineering categories

Google Drive Link - https://drive.google.com/file/d/1pWtsgA12L30BwyfxA3FJM5b3_0t4bzjn/view?usp=sharing

Popularity V/S Profitability



The correlation plot indicates a positive relationship between the popularity and profitability of menu items in the existing data. This implies that items that are popular among customers tend to contribute more to the restaurant's profits. In simpler terms, the more popular an item is, the more likely it is to generate higher profits for the restaurant. This correlation underscores the importance of understanding customer preferences and tailoring the menu offerings to meet their demands, ultimately driving profitability.

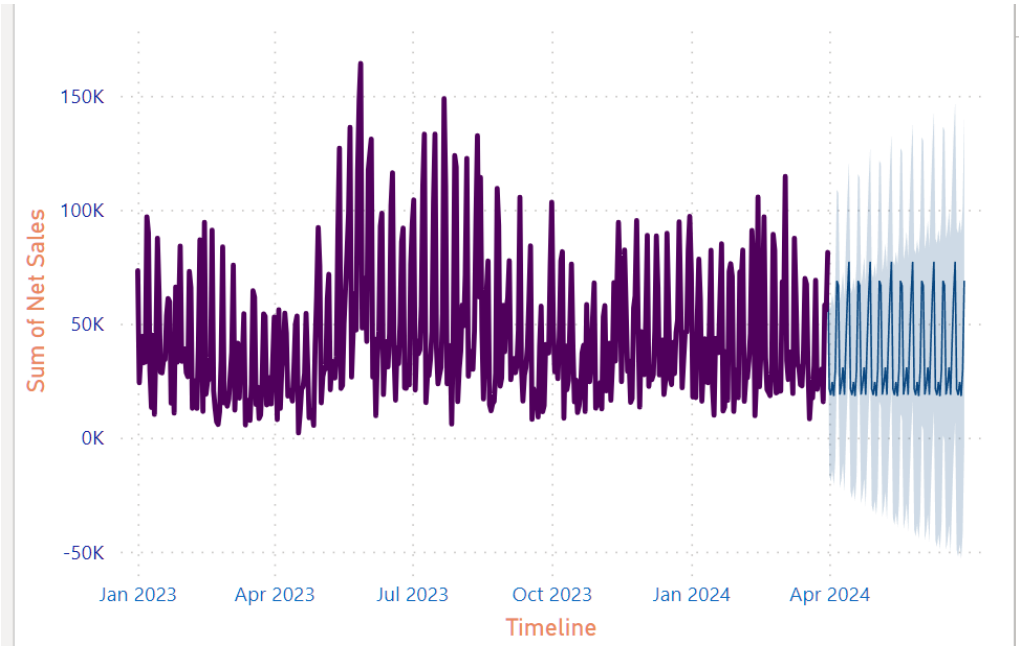
Menu Engineering Strategies:

Menu contains significant no of

- **Promotion**: Highlight star items through strategic placement, visuals, or descriptions to encourage sales.
- **Reengineering**: Adjust prices, portion sizes, or ingredients of plowhorse items to increase profitability.
- **Replacement**: Consider removing or replacing puzzle and dog items with more profitable or popular alternatives.

FORECASTING

Applied two main techniques, regression analysis and time series analysis, to predict the sales for the months of April, May, and June in 2024. Regression analysis allowed us to model the relationship between sales and various predictor variables, while time series analysis helped us capture the temporal



patterns and trends in the sales data over time. By combining these methods, we were able to generate forecasts for future sales.

Additionally, to account for uncertainty in the forecasts, we calculated the **possible range of sales** using a 95% confidence interval. This range indicates the range within which we are confident that the actual sales amounts will fall, considering the variability and potential fluctuations in sales data.

Months	Predicted Sales in Amount	Possible Range of Sales
April 2024	10,84,450	9,42,100 – 12,27,000
May 2024	11,16,860	9,74,470 – 12,60,000
June 2024	11,13,240	9,70,900- 12,55,620

CONCLUSION

The analysis may be subject to some degree of bias due to the unavailability or incompleteness of certain data required for a comprehensive analysis. The absence of complete data sets could potentially impact the accuracy and reliability of the conclusions drawn. There are some assumptions were made at the time of analysis.

- **Sales Patterns:** The analysis revealed distinct peak sales hours during the afternoon and evening, with weekends exhibiting higher sales volumes compared to weekdays. Seasonal variations were also observed, with sales peaking in May and declining in September.
- **Cost Management**: The report identified non-chargeable invoice types as significant contributors to costs, emphasizing the importance of implementing cost management strategies to improve profitability.
- **Menu Optimization**: Menu price needs to be optimized according to the markets standards and categorized according to the menu engineering factors.
- **Menu Organization:** Inadequate menu categorization was identified as a challenge, impacting customer experience and potentially leading to missed sales opportunities.
- Based on the **exploratory data analysis** performed, it was observed that customers tend to prefer food items that are moderately priced. This indicates a preference for items that strike a balance between

affordability and perceived value, avoiding extremes of being overly cheap or excessively expensive.

- After dividing **unit Rate** into three different **Price Range** categories, we conclude that nearly 54% items come under medium price range and rest of the item falls under the categories high and low (24.4%,21.1% respectively)
- Most of the items falls under star category (high Popularity, high Profitability) So, here we have advantage of making some % of changes in prices these items.

RECOMMENDATIONS

- Prices for prime dishes can be hiked in a appropriate percentage but the prices for items like starters, desserts, soups, needs a little attention. prices of these items can be reduced to retain the customer.
- **Proper customer feedback should be recorded for understanding the likes and dislikes of customers. Knowing your customer choices is the biggest weapon for predictive analysis.**
- Doing a menu engineering analysis twice a year can help you boost your sales, some menu engineering analysis is always better than none. If this activity is included on regular basis, then we can identify the best portion size and best range of prices for each Category.
- **Combo Deals and Special Offers:** The items fall under the Dog category which are not so popular and profitable should be either promoted in the form of combinations or should be removed.
- To address the issues related to plowhorse and puzzle menu items, strategies can be implemented to improve their performance and overall profitability. Introduction of combos of puzzles/ plowhorse items with star items are suggested.
- Offer or discounts should be introduced to maximize sales during weekdays.